

.PARAM

+ UN_verkettet=20000
+ IN=14.434k
+ Pel=400e6
+ fN=50Hz
+ Kc=0.5
+ UE0=600V
+ IE0=2000A
+ p=2
+ Streuung=0.05
+ rcu=0.01
+ rbg=0.001
+ Schenkelpfaktor=1 ; cylindrical pole
+ Daempferfaktor=1 ; damping factor
+ err0=0
+ n0=1500
+ J=100000
+ a=0.9 m=11 ; newer a=0.6 m=7
+ Set a,m for nonlinear noload line

.PARAM

+ UN={UN_verkettet/sqrt(3)}
+ ZN={ZN/IN}
+ Rs={ZN*rcu}
+ omega_el={2*pi*fN}
+ g={IE0/((1+Streuung)*Kc*IN*sqrt(2))}
+ Ls={Streuung*ZN/Kc/omega_el}
+ PsiN={sqrt(2)*UN/omega_el}
+ RE={UE0/IE0}
+ RD={Rs/Daempferfaktor}
+ ImNx={IE0/g}
+ ImNy={ImNx/Schenkelpfaktor}
+ MN={Pel*(1-rbg-rcu)*p/omega_el}

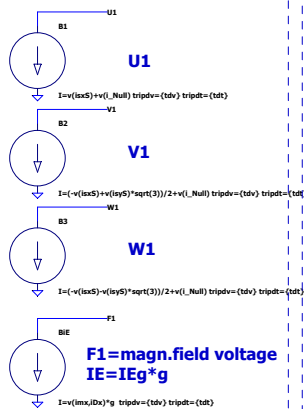
Simulate with
LTspice Version
24.0.12 !!

.options vntol=1m abstol=1m reltol=0.003

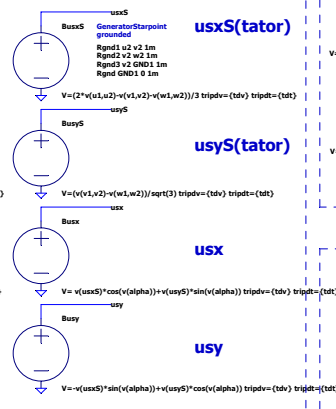
.tran 0 5s 0 0.1m

.param tdv=1 tdt=1 fn1=50 omega_n={2*pi*fn1}

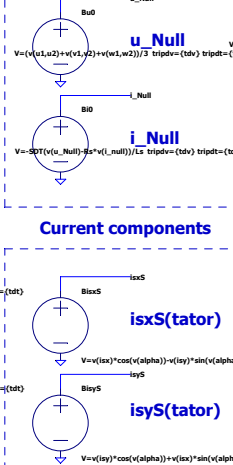
Winding current/Generator Terminal



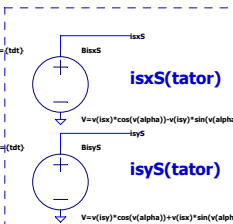
Voltage components



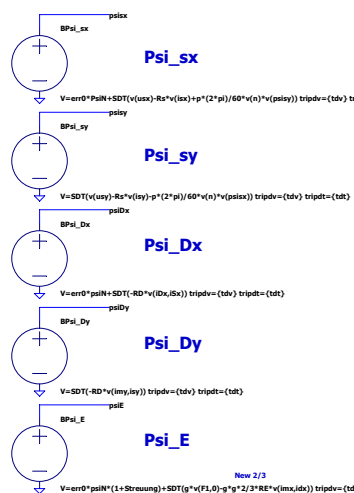
Zerosystem



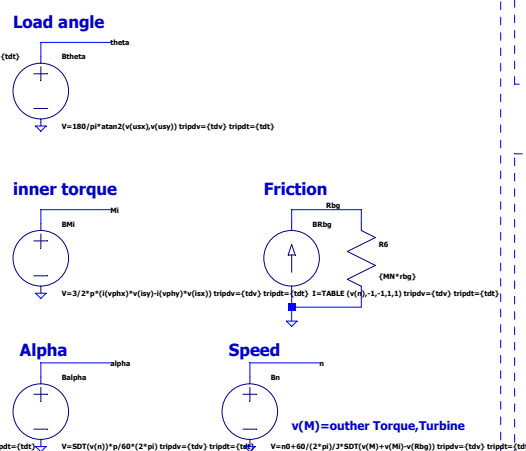
Current components



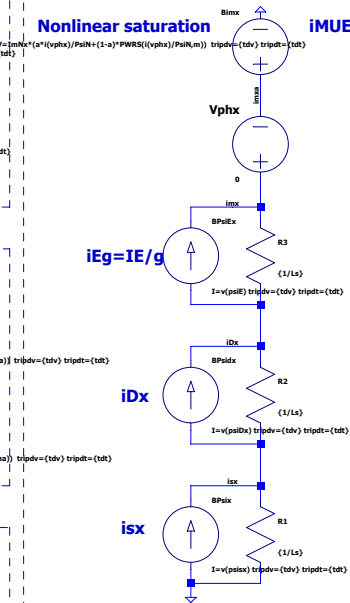
Fluxcomponents (Rotory)



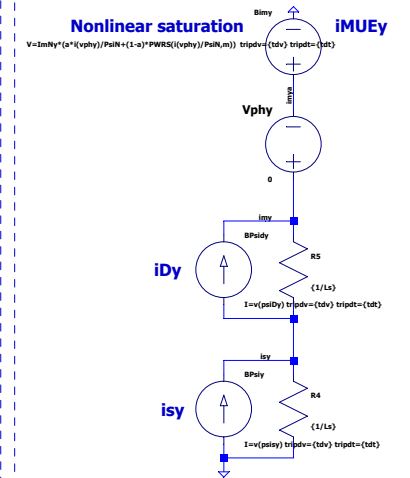
Mechanical components



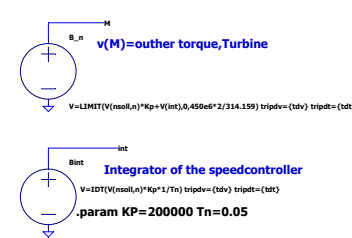
magnetic coupling direct axis with saturation isx, idx, iE (Rotory)



magnetic coupling q-axis with saturation isy, idy (Rotory)



Speed controller and magnetic field voltage



Fixed magnetic field voltage
Verr f1 0 pwl 0s 600

Setpoint of the speed controller
Vnsoll nsoll 0 pwl 0s 1500

Loadstep after 3.5s to 10MW

Vsw ctrl 0 PwL 0s 0v 3.5s 0v 3.51s 1v
S1 U1 U1a ctrl 0 MS1
S2 V1 V1a ctrl 0 MS1
S3 W1 W1a ctrl 0 MS1
.model MS1 SW(Ron=1e-3 Roff=1e10
+ Vt=0.9 Vh=0 Lser=0 Vser=0)
R11 U1a stp 40
R12 V1a stp 40
R13 W1a stp 40
Rgnd4 stp 0 1e12